5. 280mm Continuous Finishing Mill. (F-11, F-12, F-13 & F14)

5.1 Reduction cum Pinion Gear Box — 4 Nos.

Combined Gear-Cum-2 High Pinion type designed according to DIN Standards.

Normal Rating	:	300 KW
RPM	:	1000-1800
Service Factor	:	2.5 (On Load)
Duty	:	Rolling Mill
Bearings	:	SPH. Roller Bearings

Materials of construction are as follows:

- Gears & Pinions will be of forged 18CrNiMob-7/case Carburized steel hardened to 60±2 HRC, shrunk fitted on En-9 forged steel shafts. The gears and pinions will conform to DIN 3 / 4 Standard.
- Helical teeth hardened and Profile Ground Finish on Gear grinder.
- Shafts Heat Treated Steel.
- Pinion shaft mounted on anti-friction bearings.
- Mechanical Seals will be provided to avoid oil Leakage.
- Oil seal provided to stop oil leakage and ingress of water into the housing.
- Gears enclosed in Steel Fabricated housing, duly stress relieved & machined to close tolerance.
- Transparent window provided on both sides of the housing to observe teeth and lubrication.
- Bolts: High Tensile Steel
- Gearboxes will be complete with internal and external piping suitable for central oil lubrication at a supply pressure of 0.8 to 1.5 bar. An isolating valve, a Y-Strainers, Pressure Gauge, Pressure switch and a Flow Switch included in supply line.

5.2 280mm Mill Stands **4 Nos.**

Type : 2 Hi
Size : 280MM
Roll Dia. : Max. 310 MM
Min. 260 MM
Barrel : 500 MM (or as per requirement)
Roll Neck Bearing : Roller Bearing
Housing & Cap : Cast Steel – Grade-2 (IS-1030)
All Fasteners : High Tensile Steel Gr.8.8 min.
Material for Chocks : Cast Steel with 500-600 N/mm2 U.T.S.
Chocks are accurately machined and are inter changeable.
Top Roll Adjustment : By screw down mechanism.

- The housing shall be accurately machined on CNC machine and fitted with wear plates by high tensile Allen Cap Screw. This ensures accurate fitting of chocks in housings. The wear plates are ground Finished
- The cap shall be fixed over the housings by means of two tapper wedges which gives very good rigidity and does not loose during Mill operation.
- Mill Stand will be complete with all fitments, with Cooling Water piping and valves, roll cooling and Guide Cooling and Foundation bolts etc.
- The accurate machining of parts and housings keep vibration level to minimum during rolling.

5.3 Cardan Shafts **8 Nos.**

Location : Between Pinion Stand and rolls
Type : Cross Joint Type
Material of coupling heads : EN-9 (Forged)

CONSTRUCTION

- Cardan Shaft & Couplings are be required to connect output shafts of Pinion Stand to rolls of mill stand.
- Each set shall consist of two couplings hubs assemblies and one cardan Shaft.
- Cardan Shaft will be selected for the required duties from renowned manufacturers.
- The connecting Flange shall have keys to ensure positive transmission and reduce the loads on bolts.
- Connecting Hub suitable for roll end on one side & pinion end on the other side are provided.

5.4 Geared Coupling **4 Nos.**

Type	:	full flexible Geared Coupling to be connected between Reduction Gear Box, Pinion Stand and Motor

Design Parameters

Normal Rating	:	300 kW
RPM	:	1000 - 1800
Service Factor	:	≥ 3
Type of Drive	:	Non reversible for all
Duty	:	Rolling Mill Duty

5.5 Foundation Bolts **1 Set**

5.6 Motor Base Plate **16 Nos.**